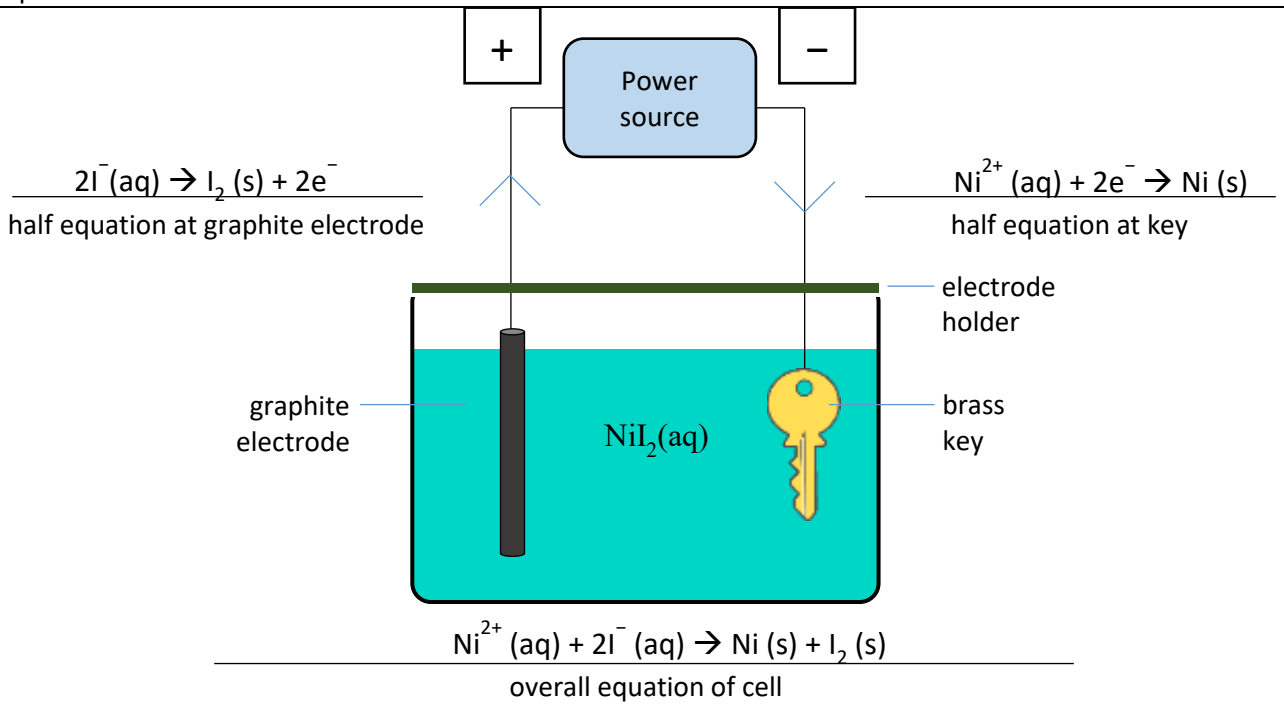
<table>
<tr><th>Q</th><th>A</th><th>Comments</th></tr>
<tr><td>1 a. i</td><td>$K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$✓</td><td></td></tr>
<tr><td>1 a. ii</td><td><table>
<tr><td></td><td>SO_2</td><td>O_2</td><td>SO_3</td></tr>
<tr><td>I</td><td>4</td><td>4</td><td>0</td></tr>
<tr><td>C</td><td>−3.2</td><td>−1.6</td><td>+3.2</td></tr>
<tr><td>E</td><td>0.8</td><td>2.4</td><td>3.2</td></tr>
</table>✓</td><td>1 mark for all three equilibrium amounts</td></tr>
<tr><td>1 a. iii</td><td>$[SO_2]$ = 0.4 M, $[O_2]$ = 1.2 M, $[SO_3]$ = 1.6 M✓
K_c «= $\frac{1.6^2}{0.4^2 \times 1.2}$ »= 13 M^{-1}✓</td><td>1 mark for all three equilibrium concentrations
1 mark for final answer with correct units
Allow ECF for M2 if equilibrium expression incorrect.</td></tr>
<tr><td>1 b. i</td><td>System is at equilibrium✓</td><td>Accept “rate of production is equal to rate of consumption”</td></tr>
<tr><td>1 b. ii</td><td>10 minutes: catalyst added OR temperature increased✓
20 minutes: volume decreased✓
30 minutes: additional oxygen added✓</td><td>Do not accept “pressure increased” for 20 min</td></tr>
<tr><td>2 a.</td><td>$2CO(g) + 2NO(g) \rightarrow 2CO_2(g) + N_2(g)$✓</td><td></td></tr>
<tr><td>2 b.</td><td>CO is a toxic gas✓</td><td></td></tr>
<tr><td>2 c.</td><td>Increased surface area of the catalyst will increase catalyst efficiency✓</td><td></td></tr>
<tr><td>2 d.</td><td>«As temperature increases» the proportion of particles with $E>E_a$ increases, and✓
the frequency of collisions with the catalyst increases. ✓</td><td></td></tr>
<tr><td>3 a. i</td><td>K^+ (l) + e^- → K (l) ✓</td><td>Do not accept (aq)</td></tr>
<tr><td>3 a. ii</td><td>n(e^-) «= 0.152 ÷ 39.1» = 3.89 × 10^{-3} mol✓
Q «= n(e^-) × F» = 375 C✓
I «= 375 ÷ 60» = 6.25 A✓</td><td></td></tr>
<tr><td>3 b.</td><td><table>
<tr><td>Electrode</td><td>Product(s)</td></tr>
<tr><td>Anode</td><td>O_2✓</td></tr>
<tr><td>Cathode</td><td>H_2✓</td></tr>
</table></td><td>Ignore states
Ignore H_2O for M1 if second
Ignore OH^- for M2 if second</td></tr>
<tr><td>4 a.</td><td>«Q = 2.00 × 6.75 × 60 =» 810 C✓</td><td></td></tr>
<tr><td>4 b.</td><td>n(e^-) «= 810 ÷ 96500» = 0.00839 mol✓</td><td></td></tr>
</table>

4 c.	n(Au) «= 0.551 ÷ 197.0» = 0.00283 mol✓ «n(e^-) ÷ n(Au) = 2.97, therefore» charge is 3+✓	
5 a.	*Lines ruled at correct position on graph*✓ Q = 150 C✓	Lines must be ruled accurately Allow ECF for mark 2 for ruled but non square lines
5 b.	n(Cu) «= 0.048 ÷ 63.5» = 7.6×10^{-4} mol✓ Q «= 150 ÷ 7.6×10^{-4}» = 2.0×10^{6} C✓ OR $Q_M = \frac{Q \times M(Cu)}{0.048}$✓ = 2.0×10^{6} C✓	
5 c.	«n(e^-) = 2n(Cu)» F «= 2.0×10^{5} ÷ 2» = 1.0×10^{5} C mol^{-1}✓	
5 d.	One of: copper not adhered to cathode side reactions impurities with lower molar mass✓	Do not accept copper still wet (causes lower F value)
6 a-d		

<table>
<tr><td>7 a.</td><td>
<table>
<tr><td></td><td>Equilibrium yield of NH_3
increase, decrease, unchanged</td><td>Equilibrium constant, K
increase, decrease, unchanged</td><td>Rate of reaction
increase, decrease, unchanged</td></tr>
<tr><td>Temperature increased to 550°C</td><td>Decrease✓</td><td>Decrease✓</td><td>Increase✓</td></tr>
<tr><td>Pressure increased to 500 kPa</td><td>Increase✓</td><td>Unchanged✓</td><td>Increase✓</td></tr>
</table>
</td><td></td></tr>
<tr><td>7 b.</td><td>No effect✓</td><td></td></tr>
<tr><td>7 c.</td><td>Catalyst provides an alternate reaction pathway with lower E_a✓
More particles will have $E \geq E_a$✓
Leading to a higher rate of successful collisions✓</td><td>Accept “rate of reaction” or “proportion of successful collisions” for “rate of successful collisions”
Ignore references to direction of reaction</td></tr>
<tr><td>7 d.</td><td>The system will act to reduce the concentration of N_2✓
By favouring the forward reaction «producing more NH_3»✓</td><td></td></tr>
<tr><td>8 a.</td><td>Any two of:
decrease in the mass of flask + contents
increase in pH
increase in volume of gas collected
time taken for solid to completely dissolve</td><td></td></tr>
<tr><td>8 b.</td><td>The rate of the reaction decreases with time✓
because the concentration of the acid decreases✓</td><td>Accept SA decreases as $CaCO_3$ is used up</td></tr>
<tr><td>9 a.</td><td>$Cr_2O_7^{2-}$ (aq) + $14H^+$ (aq) + $6e^-$ → $2Cr^{3+}$ (aq) + $7H_2O$ (l)
Correctly balanced reactants and products✓
Correct states✓</td><td></td></tr>
<tr><td>9 b.</td><td>Ag deposited: D✓
Pb deposited: E✓
Cr deposited: A✓</td><td></td></tr>
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